

Takotsubo syndrome: more frequent in women, more dangerous in men.

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Takotsubo syndrome (TTS) is an acute myocardial disease characterized by reversible left ventricular dysfunction, in the absence of obstructive coronary artery disease, caused by adrenergic overactivity and associated with non-negligible morbidity and mortality. Takotsubo syndrome, by far more frequent in women, who account for 9 out of 10 cases, is generally triggered by intense psychoemotional stress. In men, TTS has different, though not yet fully defined, characteristics and clinical course. In fact, men have a higher prevalence of a physical trigger and comorbidities, such as bronchopulmonary or cerebral pathologies, diabetes mellitus, and malignant neoplasms. The hospital course is burdened by a higher rate of cardiogenic shock and mortality. The long-term prognosis is also less favourable in men. Takotsubo syndrome in men characterizes a higher-risk phenotype, which requires close monitoring during hospitalization and careful surveillance during follow-up.

Takotsubo or stress cardiomyopathy: role of nuclear cardiology using (123)I-MIBG.

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Takotsubo cardiomyopathy, also known as broken heart syndrome or stress cardiomyopathy, is a very interesting syndrome of acute transient left ventricular dysfunction, usually following significant emotional stress. It was first described in Japan nearly two decades ago and many aspects of its pathogenesis still remain poorly understood. The incidence of out-of-hospital sudden death related to Takotsubo is currently unknown. Excess catecholamines following stress seem to trigger Takotsubo and play an important role. The clinical presentation resembles acute myocardial infarction, including chest tightness and/or dyspnea, ECG changes and elevated cardiac enzymes. However, in contrast to a typical acute myocardial infarction, no significant coronary lesions or thrombi are found on coronary angiography. Differentiating Takotsubo from acute myocardial infarction is important to avoid the unnecessary risks of thrombolytic therapy. Typically, left ventriculography shows marked abnormalities with akinesia in the mid-distal anterior wall and apex (occasionally involving other heart regions), giving a balloon shape to the left ventricle. The name Takotsubo originates from the shape of the left ventricle, which resembles a Japanese octopus-trapping pot. Hospital mortality is low but death can be caused by severe acute heart failure and/or ventricular arrhythmias. Typically, a stressful life event is reported preceding the acute symptoms. Takotsubo is most common in menopausal women although young individuals, including men, can also be affected. The autonomic nervous system has a defined role in the process. In this article, we will review the role of imaging the heart using (123)I-meta-iodobenzylguanidine, a radioactive marker allowing mapping of the autonomic nervous system of the heart, in cases of suspected Takotsubo.

Takotsubo syndrome (stress cardiomyopathy): an intriguing clinical condition in search of its identity.

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Takotsubo syndrome is a relatively frequent clinical entity presenting typically as an acute coronary syndrome in the absence of obstructive coronary artery disease and characterized angiographically by

transient left ventricular systolic dysfunction, sparing the basal segments of the left ventricle ("apical ballooning"). Takotsubo syndrome characteristically affects peri- or postmenopausal women, albeit recent series show that men also are at risk. Takotsubo syndrome is characteristically triggered by severe emotional or physical stress, which suggests a pathogenic role for increased sympathetic activity leading to myocardial perfusion abnormalities and ventricular dysfunction. The reasons why severe emotional and physical stress result in the development of takotsubo syndrome in certain individuals but not others is still a matter of speculation, but strongly suggests the existence of predisposing factors/mechanisms in certain subjects. The present article reviews the different factors that can play a role in the development of takotsubo syndrome in different patients. We propose that triggers (ie, emotional stressors, physical stressors, iatrogenic stressors, and neurologic triggers), pathogenic mechanisms (ie, increased catecholamine levels, coronary vasomotor abnormalities leading to myocardial ischemia), and predisposing factors (ie, cardiovascular risk factors, endothelial dysfunction, comorbidities) all interact in a complex fashion and possibly differently in different patients to cause takotsubo syndrome. Identifying these factors may help in preventing and managing the condition more effectively.

Takotsubo cardiomyopathy after delivery in an oestrogen-deficient patient.

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Takotsubo cardiomyopathy is characterised by a reversible left ventricular wall motion abnormality that is observed as apical ballooning without significant coronary arterial stenosis; it occurs predominantly in postmenopausal women. Here, we report a case of Takotsubo cardiomyopathy after delivery in a patient with oestrogen deficiency due to Turner's syndrome. This case shows the pathogenic role of an inadequate oestrogen level in Takotsubo cardiomyopathy.

Takotsubo cardiomyopathy and acute infectious diseases: a mini-review of case reports.

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Takotsubo cardiomyopathy (TTC), also defined as "stress cardiomyopathy," is characterized by a systolic dysfunction localized in the apical and medial left ventricles. Takotsubo cardiomyopathy is more prevalent in females and it is usually related to an event triggered by physical or emotional stress. We systematically explored PubMed and Embase medical information source to identify case reports showing association between infection and TTC. For each kind of infection, we collected a set of data, including pathogen, site of infection, clinical outcome, patient age and sex, and author and year of publication. We found 26 articles dealing with 27 case reports (74% women). The mean age was 61.4 ± 13.7 years and bacterial infections were more frequent ($n = 23$, 85.2%). In 14 cases, there was a culture-based definition of the bacterial strain: gram+ in 8 cases (57.1%) and gram- in 6 cases (42.9%). Clinical outcome was always favorable.

Reduced estrogen in menopause may predispose women to takotsubo cardiomyopathy.

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Takotsubo cardiomyopathy (apical ballooning syndrome) has been reported with increased frequency, most commonly in postmenopausal women. Despite the gender disparity, no clear link between estrogen and its possible cardioprotective effects has been shown. We present a case series of takotsubo cardiomyopathy in women and examine the prevalence of estrogen replacement therapy (ERT), in addition to conducting a systematic literature review on this topic. Consecutive cases of takotsubo cardiomyopathy were identified at our institution, Cleveland Clinic Florida, from January 2006 to December 2008, and patient-level data were extracted for analysis. For the literature review, we searched the MEDLINE database from January 1990 to March 2008 for English-language publications, using the terms apical ballooning syndrome, takotsubo, and stress cardiomyopathy, and identified case reports and series of takotsubo cardiomyopathy. Articles describing female patients and their medication use at time of presentation were included in the study. Eighteen cases of takotsubo cardiomyopathy were identified at our institution, all in postmenopausal women except for 2 who were still menstruating. Of the 16 postmenopausal cases, none were taking ERT at time of presentation. From the literature review, >400 publications were queried, of which 296 were recognized as case reports or series, with 7 articles meeting all of our inclusion criteria. From these reports, 13 women were identified, none of whom were taking ERT at time of presentation. Lack of estrogen replacement in the postmenopausal state may predispose women to takotsubo cardiomyopathy. Further studies are needed to establish the link more firmly.

A possible relationship between Takotsubo cardiomyopathy and female sex steroid-related modulation of functional cerebral asymmetry.

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Takotsubo cardiomyopathy (Tc) is a transient left ventricular apical ballooning syndrome, with symptoms and signs of acute myocardial infarction. Tc syndrome, which occurs predominantly in postmenopausal women, is characterized by increase of sympathetic activity. Studies on the gender-specific differences in sympatho-vagal regulation and functional cerebral asymmetry (FCA) imply that female pattern of dominance is characterized by the left hemisphere, which is believed to have parasympathetic predominance, whereas male pattern indicates dominance of the right hemisphere, which is believed to have sympathetic predominance. Fluctuating levels of female sex steroids are supposed to change FCA, modulating transcallosal inter-hemispheric inhibition across the menstrual cycle. The findings suggest that FCA is enhanced during the low steroid phase (menstrual phase), whereas, during high estrogen and/or progesterone phases (follicular and luteal phase) FCA is reduced. This theory is in line with concept of decreased magnitude of inter-hemispheric cortical lateralization in premenopausal women compared to men and postmenopausal women. Therefore, if postmenopausal women are more lateralized for a variety of cerebral functions, they have less balanced equilibrium between the right-sided sympathetic and left-sided parasympathetic predominance. Decrease of endogenous female sex steroid levels in postmenopausal women leads to reduced influence of estrogens to the left hemisphere, which is believed to have parasympathetic predominance. If both of these mechanisms result in sympatho-vagal imbalance, increasing sympathetic system activity in postmenopausal women, it seems reasonable why postmenopausal women became more susceptible to sympathetically-mediated syndromes such as Takotsubo cardiomyopathy.

The sex-dependent response to psychosocial stress and ischaemic heart disease.

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Stress is an important risk factor for modern chronic diseases, with distinct influences in males and females. The sex specificity of the mammalian stress response contributes to the sex-dependent development and impacts of coronary artery disease (CAD). Compared to men, women appear to have greater susceptibility to chronic forms of psychosocial stress, extending beyond an increased incidence of mood disorders to include a 2- to 4-fold higher risk of stress-dependent myocardial infarction in women, and up to 10-fold higher risk of Takotsubo syndrome—a stress-dependent coronary-myocardial disorder most prevalent in post-menopausal women. Sex differences arise at all levels of the stress response: from initial perception of stress to behavioural, cognitive, and affective responses and longer-term disease outcomes. These fundamental differences involve interactions between chromosomal and gonadal determinants, (mal)adaptive epigenetic modulation across the lifespan (particularly in early life), and the extrinsic influences of socio-cultural, economic, and environmental factors. Pre-clinical investigations of biological mechanisms support distinct early life programming and a heightened corticolimbic-noradrenaline-neuroinflammatory reactivity in females vs. males, among implicated determinants of the chronic stress response. Unravelling the intrinsic molecular, cellular and systems biological basis of these differences, and their interactions with external lifestyle/socio-cultural determinants, can guide preventative and therapeutic strategies to better target coronary heart disease in a tailored sex-specific manner.

[The broken heart syndrome].

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Tako-tsubo syndrome is characterized by acute myocardial depression secondary to psychological stress. The syndrome is ten times more common in females compared to males and is manifested by chest pain and dyspnea. ECG changes are similar to those seen in acute myocardial infarct. However, in this syndrome, cardiac enzyme levels are normal and coronary blood flow is not compromised. The syndrome is caused by excess of catecholamines. The prognosis is relatively favorable with full recovery in most patients. The recurrence rate is between 3 to 8% and there is no advantage in favor of specific treatment. This report reviews the recent literature and presents a patient who suffered from the syndrome.

In-hospital mortality among patients with takotsubo cardiomyopathy: a study of the National Inpatient Sample 2008 to 2009.

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Takotsubo cardiomyopathy is characterized by acute, reversible left ventricular apical ballooning. Little is known about the characteristics of patients with takotsubo cardiomyopathy who have in-hospital mortality. We sought to determine in-hospital mortality rate, complication rate, and characteristics of patients with in-hospital mortality related to takotsubo cardiomyopathy. Patients diagnosed with takotsubo cardiomyopathy in the National Inpatient Database Samples 2008 to 2009 using International Classification of Diseases, Ninth Revision, code 42983 were included in this study. Our primary outcome was in-hospital mortality. In patients with takotsubo cardiomyopathy, we assessed demographic factors, the prevalence and associated mortality of underlying critical illnesses (acute ischemic stroke, sepsis,

acute renal failure, respiratory insufficiency, and noncardiac surgery), and acute complications (acute congestive heart failure, respiratory insufficiency with congestive heart failure, cardiogenic shock, ventricular fibrillation/cardiac arrest, and intraaortic balloon pump placement). A total of 24,701 patients with takotsubo cardiomyopathy were identified. In-hospital mortality rate was 4.2%. A total of 21,994 patients (89.0%) were female. Male patients had a higher mortality rate than females (8.4% vs 3.6%, $P < .0001$). Age and race were not associated with mortality. Of patients with in-hospital mortality, 81.4% had underlying critical illnesses. Male patients with takotsubo had higher incidence of underlying critical illnesses than their female counterparts (36.6% vs 26.8%, $P = .008$). 80% of patients with in-hospital mortality. Male patients, who were significantly more likely to have underlying critical illness, had significantly higher mortality rates than female patients. The presence of underlying critical illness likely explains the higher mortality rate among male patients.
